Here A_x is the x–component of $\vec{A}$, A_y is the y–component of $\vec{A}$ and A_z is the z component of $\vec{A}$.

In a 2–dimensional Cartesian coordinate system (which is shown in the Figure 2.20) the vector $\vec{A}$ is given by

$$\vec{A} = A_x\hat{i} + A_y\hat{j}$$

If $\vec{A}$ makes an angle θ with x axis, and A_x and A_y are the components of $\vec{A}$ along x–axis and y–axis respectively, then as shown in Figure 2.21,

$$A_x = A\cos\theta,\ A_y = A\sin\theta$$

where 'A' is the magnitude (length) of the vector $\vec{A}$, $A = \sqrt{A_x^2 + A_y^2}$

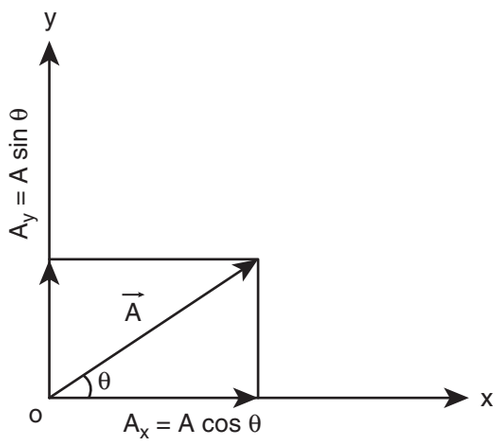


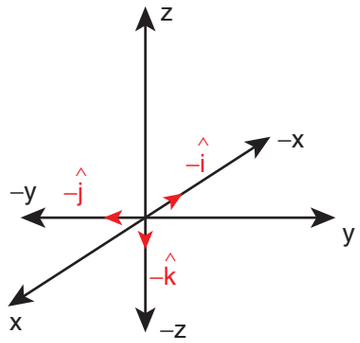
Figure 2.21 Resolution of a vector

EXAMPLE 2.3

What are the unit vectors along the negative x–direction, negative y–direction, and negative z– direction?

Solution

The unit vectors along the negative directions can be shown as in the following figure.



Then we have:

The unit vector along the negative x direction = $-\hat{i}$

The unit vector along the negative y direction = $-\hat{j}$

The unit vector along the negative z direction = $-\hat{k}$.

2.4.1 Vector addition using components

In the previous section we have learnt about addition and subtraction of two vectors using geometric methods. But once we choose a coordinate system, the addition and subtraction of vectors becomes much easier to perform.

The two vectors $\vec{A}$ and $\vec{B}$ in a Cartesian coordinate system can be expressed as

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$
$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

Then the addition of two vectors is equivalent to adding their corresponding x, y and z components.

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$$

Similarly the subtraction of two vectors is equivalent to subtracting the corresponding x, y and z components.

UNIT-2(XI-Physics_Vol-1).indd 51 05-01-2022 22:13:11